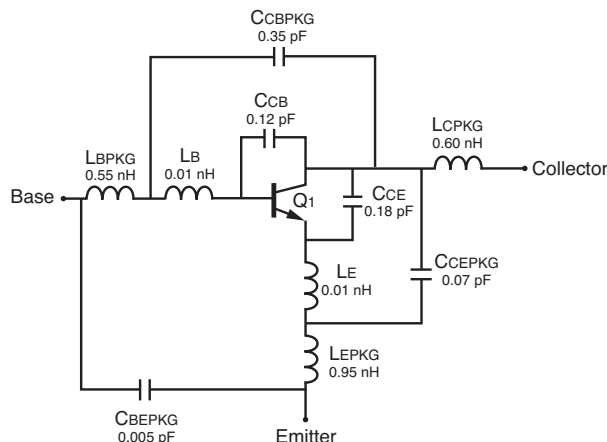


SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1	Parameters	Q1
IS	100e-18	MJC	0.76
BF	121.3	XCJC	0.5
NF	1.00	CJS	0
VA	25	VJS	0.75
IKF	293e-3	MJS	0
ISE	50e-15	FC	0.5
NE	2.111	TF	6.5e-12
BR	30	XTF	0.1
NR	1.0	VTF	10
VAR	2.105	ITF	0.5
IKR	457e-3	PTF	20
ISC	1.155e-15	TR	0
NC	1.288	EG	1.11
RE	1.5	XTB	0
RB	3.0	XTI	3
RBM	1.0	KF*	0
IRB	759e-6	AF*	1
RC	4.0		
CJE	0.6e-12		
VJE	1.049		
MJE	0.226		
CJC	83e-15		
VJC	0.89		

(1) Gummel-Poon Model

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.



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ADDITIONAL PARAMETERS

Parameters	NE894M03
CCB	0.12 pF
CCE	0.18 pF
LB	0.01 nH
LE	0.01 nH
CCBPKG	0.35 pF
CBEPKG	0.005 pF
CCEPKG	0.07 pF
LBPKG	0.55 nH
LCPKG	0.60 nH
LEPKG	0.95 nH

AF and KF are 1/f noise parameters and are bias dependent. The appropriate values for the 1/f noise parameters (AF and KF) shall be chosen from the table below, according to the desired current range.

	Ic = 5 mA	Ic = 10 mA	Ic = 15 mA
AF	1.352	2.17	2.19
KF	71.23e-15	231.3e-12	268.9e-12

For a better understanding on AF and KF parameters, please refer to AN1026.

MODEL TEST CONDITIONS

Frequency: 0.1 to 6 GHz

Bias: VCE = 1.5 V, Ic = 1 mA to 9 mA

Date: 09/2003